Customer Shopping Behavior Analysis

1. Project Overview:

This project focuses on analyzing customer shopping patterns using transaction data from 3,900 purchases across different product categories. It examines customer spending habits, purchasing preferences, customer segments, and subscription behavior to identify meaningful trends. The findings provide useful insights that can support data-driven decisions in areas such as customer retention, product planning, and business growth.

2. Dataset Summary:

- Rows: 3,900
- Columns: 18
- Key Features:
    i. Customer demographics (Age, Gender, Location, Subscription Status)
    ii. Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    iii. Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
    iv. Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python:

- **Data Loading**: Loaded the customer shopping dataset into Python using the **Pandas** library.
- **Data Exploration** : Examined the dataset structure, data types, and basic statistical information using `df.info()` and `df.describe()`.

[4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

[4]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling:** Identified missing values in the Review Rating column and filled them using the median rating calculated for each product category.
- **Column Standardization:** Renamed the dataset columns using **snake_case** to maintain a consistent and readable naming format.
- **Feature Engineering:**
    - Grouped customers into different age categories by creating an **age_group** feature.
    - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:**
    - Checked **discount_applied** and **promo_code_used** for duplicate information and removed the **promo_code_used** column.
- **Database Integration:** Established a connection between Python and PostgreSQL and stored the cleaned dataset in the database for further SQL-based analysis.

4. Data Analysis using SQL (Business Transactions)

Used **PostgreSQL** to analyze the customer transaction data and answer important business questions:

I) **Revenue by Gender**: Compared total revenue generated by male vs. female Customer.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

II) **High-Spending Discount Users**: Identified customers who used discounts but made purchases above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 839 Query complete 00:00:00.146

III) **Top-Rated Products:** Identified the top five products based on their highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

IV) **Shipping Method Comparison:** Compared the average purchase amounts for customers using Standard and Express Shipping.

	shipping_type text	Average Purchase Amounts numeric
1	Standard	58.46
2	Express	60.48

V) **Subscribers vs. Non-Subscribers:** Compared average spending and total revenue between subscribed and non-subscribed customers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

VI) **Discount Usage by Product**: Identified the top five products with the highest percentage of purchases made using discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

VII) **Customer Segmentation:** Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

VIII) **Top 3 Products per Category:** Identified the three most purchased products within each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

IX) **Repeat Buyers & Subscriptions**: Checked whether customers with >5 purchases are more likely to subscribe.

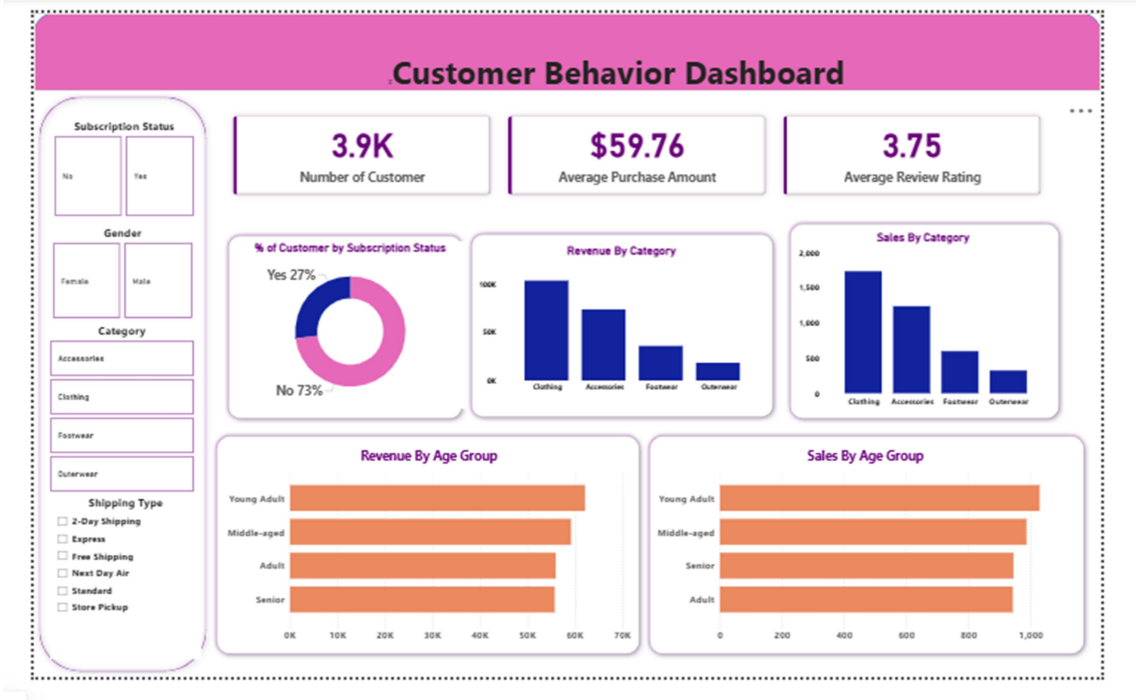
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

X) **Revenue by Age Group:** Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI:

Finally, an interactive Power BI dashboard was developed to visually present the key findings and business insights from the analysis.



6. Business Recommendations :

- **Increase Subscriptions:** Offer exclusive benefits and incentives to encourage more customers to subscribe.
- **Strengthen Customer Loyalty:** Reward repeat customers to encourage them to become loyal customers.
- **Optimize Discount Strategy:** Review discount offers to increase sales while maintaining healthy profit margins.
- **Promote Popular Products:** Use top-rated and best-selling products in marketing campaigns.
- **Target Key Customer Groups:** Focus marketing efforts on high-revenue age groups and customers using express shipping.